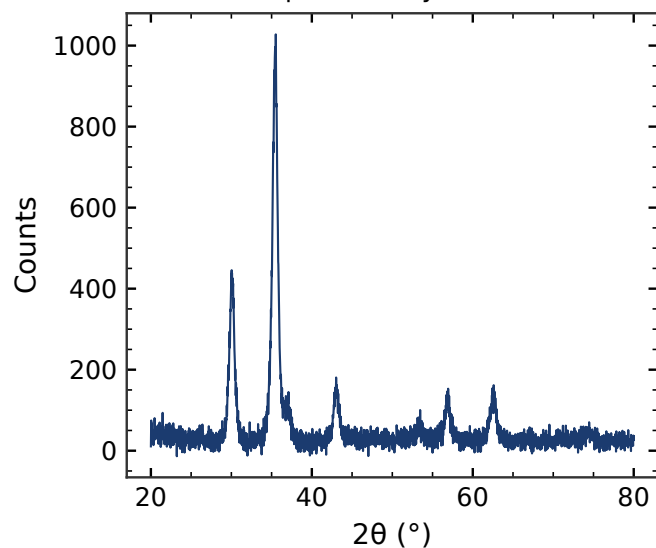


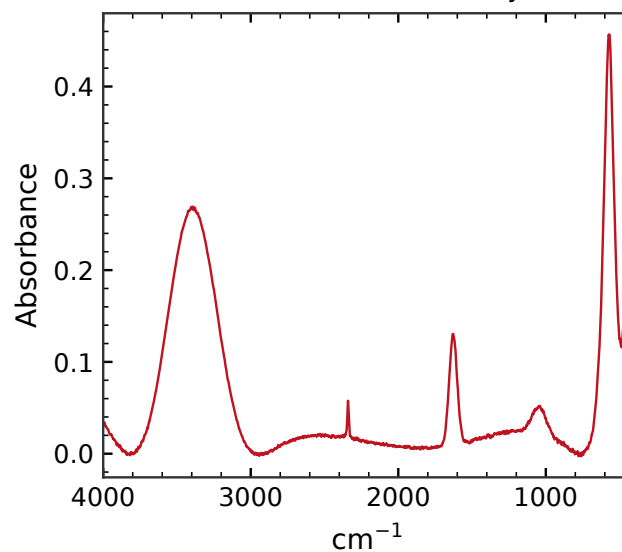
Figure 18. Cross-technique characterization summary - Fe₃O₄ nanoparticles

SYNTHETIC DEMONSTRATION DATASETS - not experimental measurements

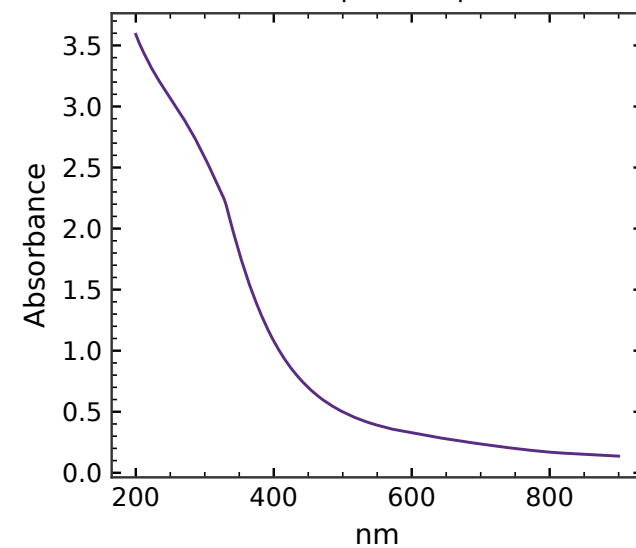
XRD - phase & crystallite size



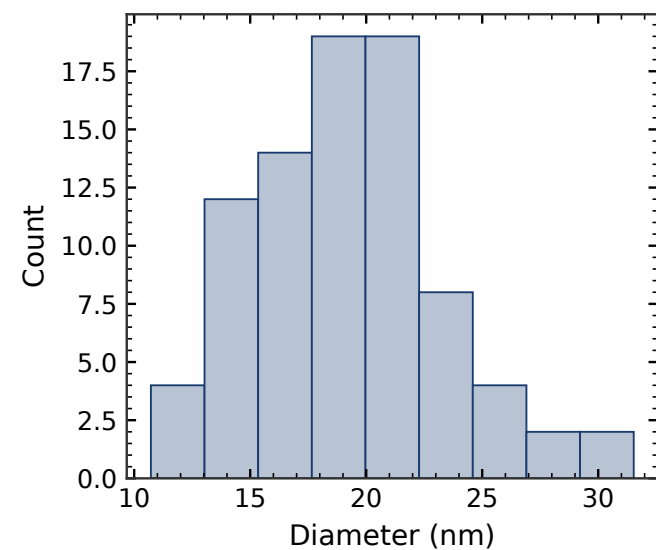
FTIR - surface chemistry



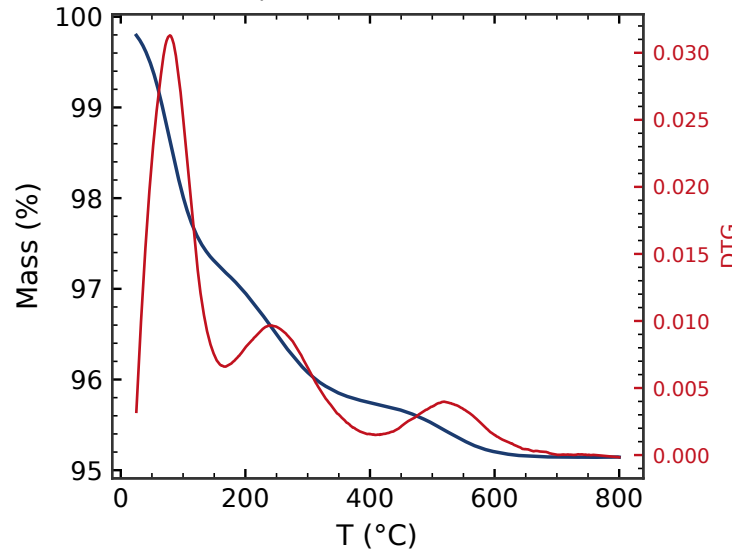
UV-Vis - optical response



SEM - size distribution



TGA / DTG - volatile content



Demonstration outcomes

Phase (XRD)	Fe ₃ O ₄ (magnetite)
Lattice parameter	8.3962 Å
Crystallite size (W-H)	11.9 nm
Microstrain	5.5e-04
SEM mean diameter	19.1 nm (n = 84)
SEM / XRD ratio	1.60
Total mass loss (TGA)	4.65 %
Optical band gap	not defensible - see report